# Learning Graph Rationales to Compress Long Chains of Thought in Multimodal Reasoning

Yizhi Wang
School of Computer Science and Engineering, Southeast University
Nanjing 210096, China
Key Laboratory of Computer Network and Information Integration (Southeast University), Ministry of Education
Nanjing 210096, China

Linan Yue*
School of Computer Science and Engineering, Southeast University
Nanjing 210096, China
Key Laboratory of Computer Network and Information Integration (Southeast University), Ministry of Education
Nanjing 210096, China

Deng-Bao Wang
School of Computer Science and Engineering, Southeast University
Nanjing 210096, China
Key Laboratory of Computer Network and Information Integration (Southeast University), Ministry of Education
Nanjing 210096, China

Tong Wei
School of Computer Science and Engineering, Southeast University
Nanjing 210096, China
Key Laboratory of Computer Network and Information Integration (Southeast University), Ministry of Education
Nanjing 210096, China

Min-Ling Zhang
School of Computer Science and Engineering, Southeast University
Nanjing 210096, China
Key Laboratory of Computer Network and Information Integration (Southeast University), Ministry of Education
Nanjing 210096, China

## Abstract

Recent advances in multimodal large reasoning models (MLRMs) have demonstrated impressive capabilities on complex multimodal tasks, yet their reliance on long Chain-of-Thoughts (CoTs) often leads to redundant reasoning and high computational cost. Existing chain-based distillation and refinement approaches alleviate redundancy to some extent but remain limited when handling ultra-long reasoning chains, suffering from weakened dependency modeling, accumulated redundancy, and the omission of critical reasoning steps. To this end, in this paper, we propose GRACE, a Graph-based Rationalized CoT Compression framework that transforms long CoTs into reasoning graphs to enable efficient CoT compression, where reasoning steps are represented as nodes and semantic dependencies as edges. Specifically, after constructing the reasoning graph, GRACE applies a graph rationale learning approach with a selector–predictor architecture. Among them, the selector estimates node-level rationale scores to sample a compact rationale subgraph, while the predictor leverages this subgraph for final prediction. The resulting subgraph serves as the distilled core rationale, retaining essential reasoning while removing irrelevant steps. Finally, the compressed CoTs are then used to build supervised fine-tuning datasets for further MLRMs tuning. Experiments on multiple multimodal benchmarks show that GRACE substantially improves inference efficiency and interpretability without sacrificing accuracy. Code is available at https://anonymous.4open.science/status/Codes-of-GRACE-AEDD.

## Keywords

Efficient Reasoning, CoT Compression, Graph Rationale Learning

**ACM Reference Format:**
Yizhi Wang, Linan Yue, Deng-Bao Wang, Tong Wei, and Min-Ling Zhang. 2026. Learning Graph Rationales to Compress Long Chains of Thought in Multimodal Reasoning. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26), August 09–13, 2026, Jeju Island, Republic of Korea.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3770855.3817627

## 1 Introduction

Multimodal Large Reasoning Models (MLRMs) [13, 24, 49, 56] are a specialized class of Multimodal Large Language Models (MLLMs). MLRMs are obtained by fine-tuning base MLLMs with long Chain-of-Thoughts (long CoTs) data through supervised fine-tuning (SFT), or by applying reinforcement fine-tuning (RFT) with reinforcement learning methods such as GRPO [32]. Building on these training strategies, MLRMs leverage structured long CoTs for reasoning and integrate self-reflection and self-correction mechanisms, which substantially improve their multimodal reasoning capabilities and decision quality.

However, during actual reasoning processes, multimodal CoTs often contain excessive intermediate steps, leading to the phenomenon of *"overthinking"* [6, 22, 29, 37], where many steps contain redundant information that does not contribute to the final answer.

*Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License.
*KDD '26, Jeju Island, Republic of Korea*
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2259-2/2026/08
https://doi.org/10.1145/3770855.3817627

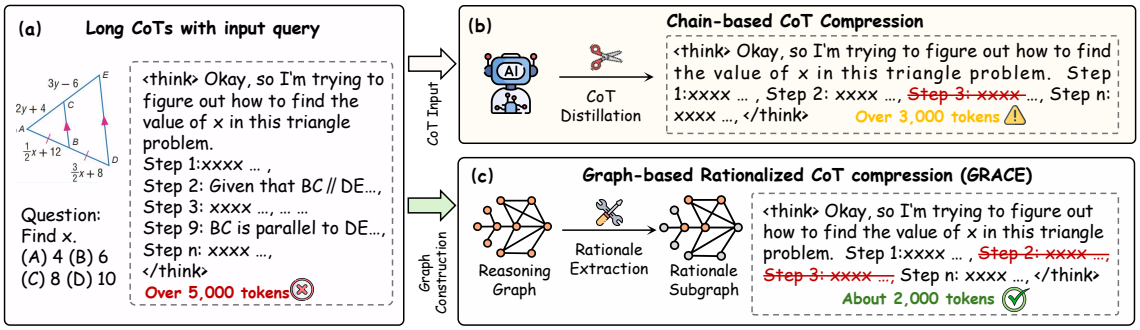



**Figure 1: Comparison between Chain-based CoT compression and our GRACE method. (a) Long CoTs generated by existing MLRMs. (b) Chain-based CoT compression, which compresses the CoT following its sequential chain structure. (c) The GRACE pipeline, which constructs a reasoning graph based on the long CoT and then extracts rationales to form the compressed CoT.**

Unlike text-only reasoning where redundancy primarily stems from verbose phrasing, multimodal CoTs exhibit structural redundancy due to the repeated verbalization of low-level perceptual cues or cross-modal correspondences. For example, as shown in Figure 1(a), given a simple problem, MLRMs restate the same visual cue for multiple times (e.g. "Given that BC || DE" and "BC is parallel to DE") and generate an excessively long CoT to produce the answer (over 5,000 tokens). This not only increases computational cost and reduces inference efficiency but also weakens model interpretability, limiting the practical performance of MLRMs.

To mitigate this issue, a straightforward approach is to directly adapt CoT compression methods from the text domain to multimodal settings [18, 21, 47, 64]. As illustrated in Figure 1(b), existing methods predominantly adopt a chain-based structure, shortening raw multimodal reasoning traces by detecting and pruning redundant or repetitive steps, which yields more concise chains. The resulting "original query–compressed CoT" pairs are used to construct a new dataset for supervised fine-tuning (SFT). Training on this dataset could equip the model with the ability to generate more concise multimodal reasoning at inference time, thereby significantly improving its efficiency.

However, chain-based CoT compression approaches face limitations when applied to ultra-long multimodal CoTs. Multimodal reasoning trajectories typically exhibit higher redundancy and nonlinear dependencies, characterized by rich visual grounding details, spatial relationships, and vision–language alignment cues. These features give rise to cross-referential structures rather than sequential ones. Under such conditions, chain-based CoT compression struggles to adequately model the intricate interleaving dependencies. Moreover, in a linear chain, excessively long information propagation paths impair the model's ability to capture long-range contextual dependencies, often resulting in either redundant content accumulation or the omission of critical reasoning steps.

In contrast, graph structures [16, 60] can more naturally capture the dependency structure and topological relationships in multimodal reasoning, enabling comprehensive modeling of visual content and cross-modal interactions throughout the reasoning process. As illustrated in Figure 1(c), by representing each reasoning step as a graph node and semantic relationships between steps as edges, the core reasoning path and pivotal steps can be identified more accurately. This approach enhances both compression quality and reasoning efficiency in complex multimodal scenarios.

Along this line, as shown in Figure 1(c), we propose a ***G***raph-based ***Ra***tionalized ***C***oT compr***e***ssion (GRACE) to efficiently compress long reasoning chains. Specifically, we first divide the original long CoT into steps, treating each step as a graph node, and construct edges between nodes by analyzing their semantic relationships using a LLM, thereby forming a complete reasoning graph. In detail, we employ Qwen3-8B rather than flagship-scale models for reasoning graph construction, as it provides sufficient capability while remaining cost-efficient. Moreover, since the constructed graphs are used solely to derive refined SFT data and are not generated during inference, the procedure incurs only a *one-time offline cost*, resulting in minimal overall overhead. In addition, to ensure robustness and stability in graph construction, we design explicit stepwise instructions and leverage in-context learning with a diverse set of high-quality examples. Based on this constructed reasoning graph, we design the graph rationale compression framework, which consists of a *selector* and a *predictor*. Specifically, the reasoning graph and the corresponding text query are encoded by a language model (e.g., Qwen3-Embedding-8B [48]) to obtain node and query representations. The *selector* then predicts the probability of each node being a key node and samples the important subgraph accordingly, generating the corresponding subgraph representation. Finally, the subgraph representation are concatenated with the query one and fed into the *predictor* for the final prediction. After training, the *selector* can automatically identify key nodes from long CoTs, enabling efficient compression of reasoning chains. Key reasoning nodes are then merged in order and smoothed by a lightweight-LLM (Qwen3-0.5B) with only minimal edits (e.g., linking words). Ultimately, the compressed CoTs, together with the original problems, are then used to construct a new SFT dataset for further MLLMs tuning to achieve MLRMs. Through this process, the MLRMs can retain essential reasoning information while significantly reducing redundancy, improving inference efficiency and interpretability, thus achieving efficient long-chain multimodal reasoning. Extensive experiments on multiple multimodal benchmark datasets demonstrate that our method enables efficient reasoning while achieving strong performance.

In summary, our main contributions are as follows:

- We explore CoT compression for multimodal reasoning and highlight the suitability of a graph-based perspective for modeling long multimodal CoTs, as graph structures can capture structural redundancy and non-linear dependencies among reasoning steps.
- We propose GRACE, a graph-based rationalized CoT compression framework that constructs reasoning graphs from long CoTs and learns to extract compact rationale subgraphs through a selector-predictor architecture.
- We conduct extensive empirical studies on multiple multimodal reasoning benchmarks, demonstrating that GRACE substantially improves reasoning efficiency while maintaining strong accuracy and visual information preservation.

## 2 Related Work

### 2.1 Multimodal Large Reasoning Models

Multimodal Large Language Models (MLLMs) have recently achieved impressive performance across diverse tasks, including visual question answering, image captioning, and document understanding [1, 11, 14, 19]. These models integrated visual and linguistic signals within a unified generative architecture, allowing them to jointly interpret and describe complex scenes [7, 43, 51, 57]. However, while these models excelled at perception and modality alignment, they were not originally equipped with mechanisms for complex reasoning. As a result, their ability to conduct structured inference remained limited, particularly when faced with tasks requiring compositional logic or long-term dependency tracking.

Motivated by the progress of reasoning-focused language models, researchers began exploring how to extend such capabilities to multimodal scenarios, giving rise to Multimodal Large Reasoning Models (MLRMs). Early studies typically relied on explicit instruction tuning or search-based reasoning optimization to induce step-by-step reasoning behaviors. For example, Mulberry [50] and LLaVA-CoT [46] guided models to generate explicit Chains of Thought (CoTs) via carefully designed prompts or Monte Carlo Tree Search (MCTS), thereby improving transparency and interpretability in multimodal reasoning.

With the emergence of powerful text reasoning models such as DeepSeek-R1 [8, 9], DeepSeekMath [32], and other R1-style architectures, reinforcement learning (RL)–based post-training became the mainstream paradigm for enhancing complex reasoning ability [17, 30, 34]. By introducing reward-driven optimization objectives, these methods encouraged models to produce logically consistent, multi-step reasoning paths [10, 33, 39].

Among them, a line of works directly applied RL to improve multimodal reasoning, such as MM-Eureka [24] and VisualThinker-R1 [62], along with other efforts that followed a similar reinforcement-driven optimization framework [4, 27, 38, 58]. Another line of research adopted a two-stage paradigm, where models were first supervised on long, high-quality CoTs and then refined via RL post-training to enhance reasoning depth and stability. Representative examples included recent Qwen3-VL [1, 48], R1-OneVision [49], and Vision-R1 [13], as well as several related studies exploring similar hybrid strategies [5, 36]. Collectively, these approaches substantially advanced multimodal reasoning and represented a prevailing paradigm in MLRM development.

Despite their advances, current MLRMs are prone to excessive overthinking [6, 22], characterized by unnecessary reasoning paths and repeated reflection. This not only raises computational demands but also limits inference speed and interpretability, creating a significant obstacle to scaling these models for complex multimodal applications.

### 2.2 CoT Compression

Chain-of-Thought (CoT) compression aims to shorten reasoning chains while preserving core inferential information. Early approaches primarily relied on prompt-based strategies, where carefully designed prompts guided the model to produce more concise reasoning sequences [26, 31]. For instance, [12] proposed a "*Use at most k tokens*" prompt, which constrained the length of generated reasoning chains. However, such methods heavily depended on manual prompt design and were limited in scalability. More recently, researchers [35, 53] explored model-driven compression methods, which operated at different granularities of reasoning units, including token-level and step/chunk-level. After compression, these methods often used the resulting concise CoTs for SFT, helping models reduce redundancy while maintaining reasoning capability.

Specifically, *at the token level*, the focus was on evaluating the contribution of individual tokens and removing low-value tokens to simplify reasoning [44, 52, 64]. For example, [52] introduced Conditional Token Selection (CTS), which trained a reference model to dynamically estimate token importance using perplexity metrics and pruned redundant tokens. [44] proposed TokenSkip, which retained only tokens above a certain importance threshold to generate more compact CoTs. *At the step or chunk level*, reasoning chains were divided into higher-level semantic units, and only the most important steps were preserved [21]. For instance, [45] proposed the PIR (Perplexity-based Importance Refinement) framework, which evaluated the importance of progressive and functional reasoning steps using perplexity metrics and removed low-contribution steps. [41] partitioned reasoning into semantic chunks, generated multiple simplified candidates for each chunk, and used a greedy strategy to select the combination that best balanced conciseness and fidelity. Some works went further by replacing the chain structure with a graph-based representation. For example, [60] proposed Prune-on-Logic, which converted CoTs into logical graphs and pruned redundant or low-value nodes to achieve more structured compression. While these approaches showed effectiveness in text-based reasoning, research on compressing CoTs for MLRMs remained limited. Compared with text-only CoTs, multimodal reasoning chains were longer and more complex, involving interactions across different modalities, which made exploring CoT compression strategies for multimodal models particularly valuable.

### 2.3 Graph Rationale Learning

Graph rationale learning has recently attracted increasing attention as a means to enhance the interpretability of graph predictions. The core idea of these methods was to first identify key subgraphs in the input graph that were relevant to the prediction task, referred to as rationales, and then make predictions solely based on these

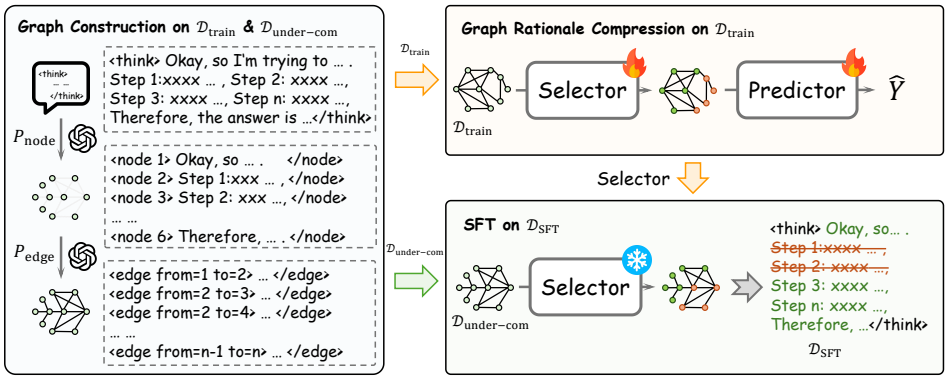



**Figure 2: The overall framework of GRACE. Both $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{under-com}}$ are first converted into graph-structured data (*Graph Construction*). Then, the selector and predictor are trained on $\mathcal{D}_{\text{train}}$ to obtain graph-based reasoning compression capability (*Graph Rationale Compression*). Finally, the trained selector is applied to $\mathcal{D}_{\text{under-com}}$ to generate the compressed dataset $\mathcal{D}_{\text{SFT}}$, which is subsequently used for supervised fine-tuning of the base model (*SFT on* $\mathcal{D}_{\text{SFT}}$).**

extracted rationales. The selected rationales thus served as direct explanations for the model's decisions. Existing studies [3, 25, 42, 54] primarily focused on improving the faithfulness and reliability of rationale extraction. For instance, [42] decomposed the input graph into rationale and non-rationale components and utilized the latter to generate counterfactual samples, enhancing the credibility of the explanations. Others [25] incorporated information bottleneck principles or attention mechanisms to constrain information flow or reduce stochasticity, ensuring that the extracted rationales accurately reflected the model's decision-making process. In this paper, we focus on the design framework of graph rationale learning, emphasizing how rationales could be leveraged for efficient compression of long reasoning chains, without delving into the specific algorithmic implementations.

## 3 Problem Definition

In this section, we formally define the objective of GRACE. Specifically, our approach consists of two stages. First, given a Graph Rationale training dataset:

$$\mathcal{D}_{\text{train}} = \{(Q_i, S_i, Y_i)\}_{i=1}^{N_{\text{train}}}, \tag{1}$$

where $Q_i = (\text{query}_i, \text{image}_i)$, $S_i = \{s_1, s_2, \ldots, s_{n_i}\}$ is a pre-generated long CoT with $s_k$ denoting the $k$-th token, and $Y_i$ is the corresponding answer, our goal is to first construct a reasoning graph $G_i$ from the long CoT $S_i$. Based on this graph, we train a Graph Rationale model consisting of a selector $\mathcal{F}_{\text{s}}$ and a predictor $\mathcal{F}_{\text{p}}$, which identifies key reasoning nodes (rationales) within the graph.

Then, for a CoT compression dataset:

$$\mathcal{D}_{\text{under-com}} = \{(Q_j, S_j, Y_j)\}_{j=1}^{N_{\text{under-com}}}, \tag{2}$$

we apply the trained selector $\mathcal{F}_{\text{s}}$ to extract the key nodes and generate a compressed CoT $S_j^c$, yielding a new supervised fine-tuning dataset:

$$\mathcal{D}_{\text{SFT}} = \{(Q_j, S_j^c, Y_j)\}_{j=1}^{N_{\text{SFT}}}, \tag{3}$$

where $N_{\text{SFT}}$ is equal to $N_{\text{under-com}}$. This SFT dataset can be used to fine-tune MLLMs to achieve MLRMs, enabling efficient compression of long-chain CoTs while retaining essential reasoning information.

## 4 Graph-based Rationalized CoT Compression

In this section, we present the implementation details of the proposed GRACE framework. The overall pipeline consists of three key components: reasoning graph construction, graph rationale compression, and SFT on the compressed data. The core idea of GRACE is to leverage the reasoning graph to automatically identify the most critical reasoning steps (rationales) from long CoTs, thereby removing redundant reasoning while preserving essential logical consistency. This process produces a compact CoT dataset for SFT, enabling more efficient and effective reasoning.

### 4.1 Reasoning Graph Construction

We begin by constructing reasoning graphs from the Graph Rationale training set $\mathcal{D}_{\text{train}} = \{(Q_i, S_i, Y_i)\}_{i=1}^{N_{\text{train}}}$. For each CoT $S_i$, the sequence is segmented into semantic units or reasoning steps, which are then modeled as nodes to form a reasoning graph $G_i = (\mathcal{V}_i, \mathcal{E}_i)$. Here, the node set $\mathcal{V}_i = \{v_{i,1}, \ldots, v_{i,|\mathcal{V}_i|}\}$ corresponds to individual CoT steps, while the edge set $\mathcal{E}_i$ encodes semantic or logical dependencies between steps.

As shown in Figure 2, to operationalize this process, we employ two prompting stages powered by a LLM. First, using a node extraction prompt $P_{\text{node}}$, each original CoT $S_i$ is decomposed into a series of semantically independent reasoning fragments, each representing a distinct inferential step. These fragments are instantiated as nodes, yielding the node set $\mathcal{V}_i$. Next, an edge extraction prompt $P_{\text{edge}}$ is applied to identify semantic and logical dependencies among the extracted nodes. For any pair of nodes $(v_p, v_q)$, the LLM determines whether a directed relation exists and records it as an edge $e \in \mathcal{E}_i$.

Through these two prompting stages, the original long CoT is transformed into a structured reasoning graph $G_i = (\mathcal{V}_i, \mathcal{E}_i)$, where each node captures a concise reasoning unit and each edge encodes explicit semantic dependencies, laying the foundation for subsequent rationale learning and compression. The prompt for node splitting is shown in Appendix D, and the prompt for edge extraction is shown in Appendix E.

***Remark.*** In the two prompting stages, we adopt Qwen3-8B instead of flagship-scale models for reasoning graph construction, as it provides sufficient capability at a much lower cost. Notably, the constructed reasoning graphs are used only to generate a compressed CoT dataset, which is then employed for subsequent SFT to enable efficient reasoning. Once generated, these reasoning graphs are reused across all downstream stages without incurring any additional inference-time overhead. Consequently, reasoning graph construction is a one-off offline process, making the overall pipeline highly cost-effective despite leveraging an LLM for structure extraction. To further ensure robustness and stability in graph construction, we design explicit stepwise instructions and employ in-context learning with a diverse set of high-quality examples.

## 4.2 Graph Rationale Compression

Given the reasoning graph $G = (\mathcal{V}, \mathcal{E})$, we introduce the graph rationale compression framework, including a *selector* and a *predictor*. Among them, the selector $\mathcal{F}_\text{s}$ is employed to automatically identify the critical reasoning nodes, and the *predictor* $\mathcal{F}_\text{p}$ yields the task results based on the identified nodes and input queries. Specifically, we first map each node text $v_j$ into an initial embedding $x_j \in \mathbb{R}^d$ using Qwen3-Embedding-8B [48], while the text query in $Q$ is encoded as $h_q \in \mathbb{R}^d$. Then, the node representations, together with the graph structure, are passed through a graph neural network (GNN) to obtain contextualized node embeddings:

$$\mathbf{H}_G = \text{GNN}(G, \{x_j\}) \in \mathbb{R}^{|\mathcal{V}| \times d}. \tag{4}$$

On top of these embeddings, the selector predicts a probability distribution over nodes being selected as rationales:

$$\tilde{\mathbf{M}} = \sigma(W_m \mathbf{H}_G), \tag{5}$$

where $W_m \in \mathbb{R}^{1\times d}$ is a trainable parameter, $\sigma(\cdot)$ is the Sigmoid function, and $\tilde{\mathbf{M}} \in (0,1)^{|\mathcal{V}|}$ represents the continuous selection probabilities.

Next, to sample a binary mask $\mathbf{M} = {m_j}_{j=1}^{|\mathcal{V}|}$ with $m_j \in \{0,1\}$ from $\tilde{\mathbf{M}}$ while ensuring differentiability, we adopt the Gumbel-Softmax trick [15]. Specifically, each $m_j$ is computed as:

$$m_j = \frac{\exp\left(\left(\log \tilde{m}_j + q_j\right)/\tau\right)}{\sum t \exp\left(\left(\log \tilde{m}_t + q_t\right)/\tau\right)}, \tag{6}$$

where $\tau$ is the temperature hyperparameter, $q_j = -\log\left(-\log\left(u_j\right)\right)$, and $u_j$ is sampled from the uniform distribution $U(0,1)$.

Based on this mask, the rationale node embeddings can be represented as:

$$\mathbf{H}_r = \mathbf{M} \odot \mathbf{H}_G, \tag{7}$$

where $\odot$ denotes element-wise multiplication. Next, a READOUT operation aggregates the rationale node embeddings into a single representation:

$$h_r = \text{READOUT}(\mathbf{H}_r). \tag{8}$$

Finally, we concatenate the rationale representation $h_r$ with the query embedding $h_q$ [1]:

$$h_z = [h_r; h_q], \tag{9}$$

and feed it into the predictor $\mathcal{F}_\text{p}$ to produce the prediction $\hat{Y} = \mathcal{F}_\text{p}(h_z)$.

By minimizing the loss between $\hat{Y}$ and the ground-truth answer $Y$, the selector $\mathcal{F}_\text{s}$ learns to extract the most representative reasoning steps, thereby compressing the original CoT. Formally, the overall loss function of GRACE can be expressed as:

$$\mathcal{L}_\text{GRACE} = L(Y, \hat{Y}) + \lambda \cdot (\eta - \frac{1}{|\mathcal{V}|}\sum_j m_j)^2, \tag{10}$$

where $L$ denotes the task-specific objective (e.g., cross-entropy loss), $\eta$ is a predefined compression ratio controlling the expected proportion of selected nodes, and $\lambda$ is a regularization coefficient.

## 4.3 Supervised Fine-tuning on Compressed CoT

After training, we can further compress long CoTs using the compression dataset $\mathcal{D}_\text{under-com} = \{(Q_j, S_j, Y_j)\}_{j=1}^{N_\text{under-com}}$. Specifically, we first convert each $S_j$ into a reasoning graph following the method introduced in Section 4.1. Then, the *selector* is applied to identify key nodes and generate compressed CoTs $S_j^c$. This process produces a new SFT dataset, denoted as $\mathcal{D}_\text{SFT}$.

Specifically, we first remove non-critical nodes from each reasoning graph according to the binary mask $\mathbf{M}$ predicted by the selector. The remaining nodes representing the selected rationale subgraph are then reassembled in their original sequential order to reconstruct a compressed CoT $S_j^c$.

However, since the selected rationale fragments may contain discontinuities or missing linguistic transitions, we employ a lightweight LLM (Qwen3-0.5B) to refine these fragments. The model is prompted to smooth the text by adding minimal connective expressions, while preserving the original reasoning semantics and structure. This refinement process ensures that the resulting $S_j^c$ remains coherent and human-readable. The final collection of $(Q_j, S_j^c, Y_j)$ pairs constitutes the supervised fine-tuning dataset $\mathcal{D}_\text{SFT}$.

We then fine-tune downstream multimodal LLMs on $\mathcal{D}_\text{SFT}$. The training objective minimizes the negative log-likelihood of generating the answer conditioned on the compressed CoT, enabling the model to produce shorter yet logically consistent reasoning chains. This process preserves the critical reasoning logic while significantly reducing redundancy, thereby improving reasoning efficiency and interpretability.

$$\mathcal{L}_\text{LLM} = -\sum_{j=1}^{N_\text{under-com}} \sum_{k=1}^{K_j} \log P_\theta\left(s_{j,k}^c \mid s_{j,<k}^c\right), \tag{11}$$

where $K_j = \text{len}(S_j^c)$, $\theta$ denotes the model parameters.

# 5 Experiments

In this section, we conduct experiments to evaluate the effectiveness and efficiency of GRACE by investigating the following research questions (RQs):

- **RQ1:** To what extent can GRACE improve model efficiency while maintaining high reasoning accuracy?
- **RQ2:** How does the compression ratio $\eta$ affect the performance of GRACE?

[1] Except for the text query, no additional modalities are introduced at this stage, since the CoT already encapsulates the understanding of other modalities, effectively converting them into text and internalizing this information within the CoT sequence.

- **RQ3:** How are the reasoning graphs constructed by GRACE characterized, and what types of nodes tend to be preserved during compression?
- **RQ4:** Could GRACE preserve the key visual information from the original reasoning trajectory?
- **RQ5:** How do models fine-tuned on GRACE-compressed reasoning data perform in terms of both efficiency and accuracy on downstream reasoning tasks?

**Table 1: Statistics of the GRACE-Dataset, where $\mathcal{D}_{\text{train}}$ is the graph rationale training data, $\mathcal{D}_{\text{under-com}}$ is the under compressed dataset, and $\mathcal{D}_{\text{SFT}}$ is the compressed dataset.**

| Item | Value |
|---|---|
| #Samples | 29,514 |
| $\lvert\mathcal{D}_{\text{train}}\rvert$ | 9,838 |
| $\lvert\mathcal{D}_{\text{under-com}}\rvert$ | 19,676 |
| Answer length of $\mathcal{D}_{\text{under-com}}$ ($\mu \pm \sigma$) | 484.6 ± 168.5 |
| Answer length of $\mathcal{D}_{\text{SFT}}$ ($\mu \pm \sigma$) | 221.3 ± 127.2 |

## 5.1 Dataset

To evaluate the influence of CoT compression on multimodal reasoning, we construct a new dataset named GRACE-Dataset, which contains 29,514 samples. This dataset integrates samples from multiple multimodal data sources, including Geo170k, RAVEN, GeomVerse, DVQA, and InterGPS, covering diverse domains such as mathematical reasoning, geometric analysis, chart understanding, commonsense question answering, and scientific reasoning. Each sample in GRACE-Dataset contains an (image, query, answer) triplet paired with a long CoT, where every query has a unique and verifiable answer. The CoTs are directly adopted from the R1-OneVision Dataset introduced in [49], where multimodal reasoning traces were generated via image–text conversion method.

As shown in Table 1, we divide GRACE-Dataset into two subsets: $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{under-com}}$. We first train the selector and predictor of GRACE on $\mathcal{D}_{\text{train}}$, and then apply the trained selector to compress the CoTs in $\mathcal{D}_{\text{under-com}}$, thereby constructing a fine-tuning dataset $\mathcal{D}_{\text{SFT}}$ for downstream multimodal reasoning models.

During evaluation, we comprehensively assess model performance on several widely recognized multimodal reasoning benchmarks, including MathVerse [59], MathVista [23], WeMath [28] and MMMU [55]. These datasets cover a broad range of visual reasoning tasks with varying complexity and modality interactions.

## 5.2 Comparison Methods

Since graph-based CoT compression remains a relatively novel and underexplored research problem, the amount of existing work in this area is still quite limited, particularly in the context of multimodal reasoning. To enable a meaningful and comprehensive comparison, we select the following three categories of representative methods as baselines:

**(1) Graph-based CoT compression methods.** We adopt Prune-on-Logic [60] as a representative graph-based CoT compression approach. Originally proposed for text-based CoT compression, it is transferred to the multimodal reasoning setting in our experiments:

- Prune-on-Logic [60]: Prune-on-Logic decomposes a CoT into two types of textual components: nodes and connectors. Nodes correspond to atomic reasoning statements, while connectors represent the logical relations between these statements. This decomposition is achieved via prompting an LLM. The method then constructs a reasoning graph by linking nodes based on semantic dependencies extracted by an LLM-based parser, and prunes the graph according to a predefined compression ratio. Node importance is evaluated and ranked based on the change in model perplexity after removing each node.

While both Prune-on-Logic and GRACE share the same overarching goal of compressing reasoning structures to improve efficiency and interpretability, their technical paradigms differ substantially. Prune-on-Logic follows a heuristic scoring approach that relies on perplexity-based importance estimation, whereas GRACE introduces a **learnable graph representation framework** capable of explicitly modeling both node-level semantics and edge-level logical relations. This design enables a data-driven and adaptive identification of critical reasoning components, thereby improving robustness and generalization beyond handcrafted heuristics.

**(2) Chain-based CoT compression methods.** We further include three representative state-of-the-art chain-based text CoT compression methods, which are transferred to the multimodal setting for comparison with GRACE:

- StepEntropy [20]: StepEntropy splits the CoT into individual reasoning steps, scores each step by summing token-level entropies derived from generation probabilities, and retains only the highest-scoring steps for fine-tuning.
- LIMOPro [45]: LIMOPro evaluates step importance based on perplexity and prunes reasoning steps with low contribution.
- R1-Compress [40]: R1-Compress chunks the reasoning process and searches for optimal combinations of steps to achieve compression.

**(3) Multimodal Large Reasoning Models.** Beyond reasoning compression methods, we also benchmark GRACE against several representative multimodal reasoning models:

- No Compression: Fine-tune the base model using the original, uncompressed CoTs (i.e., $\mathcal{D}_{\text{under-com}}$) from the GRACE-Dataset.
- R1-OneVision [49]: R1-OneVision enhances step-by-step reasoning capability through supervised fine-tuning on long CoT annotations.
- Vision-R1 / Vision-R1-CI [13]: Vision-R1-CI is supervised fine-tuned on long CoT data to improve reasoning ability, and Vision-R1 is further obtained via an additional reinforcement learning–based post-training stage, which enhances reasoning depth.

It is worth noting that all of the above models belong to the category of long CoT reasoning models, which typically suffer from varying degrees of overthinking. Moreover, we do not report the results of Vision-R1 and Vision-R1-CI on the WeMath dataset, as this dataset is included in their training corpus, making direct comparison unfair. This setup ensures that all reported results reflect fair and unbiased evaluations across all benchmarks.

## 5.3 Experimental Setup

We adopt the Qwen2.5-VL series as the base model for all experiments, considering its strong multimodal understanding capability and compatibility with large-scale visual–language reasoning tasks.

**Table 2: Performance comparison on three multimodal reasoning benchmarks. Accuracy (Acc.) measures effectiveness. Average CoT length (Avg. length) reflects reasoning efficiency. Bold indicates the best performance among CoT compression methods.**

| Method | MathVerse | | MathVista | | WeMath | | MMMU | |
|---|---|---|---|---|---|---|---|---|
| | Acc. (%) ↑ | Avg. length (#tok) ↓ | Acc. (%) ↑ | Avg. length (#tok) ↓ | Acc. (%) ↑ | Avg. length (#tok) ↓ | Acc. (%) ↑ | Avg. length (#tok) ↓ |
| Vision-R1 | 43.6 | 639.1 | 68.9 | 482.1 | - | - | 49.3 | 722.5 |
| R1-Onevision | 43.0 | 578.9 | 67.3 | 414.4 | 58.9 | 548.4 | 48.7 | 649.8 |
| Vision-R1-CI | 41.8 | 598.9 | 67.5 | 358.2 | - | - | 47.8 | 654.3 |
| No Compression | 42.7 | 504.8 | 67.5 | 342.9 | 59.5 | 492.0 | 48.4 | 594.2 |
| LIMOPro | 40.5 | 432.7 | 66.4 | 329.8 | 58.1 | 435.1 | 47.3 | 492.0 |
| R1-Compress | 38.8 | 368.6 | 64.3 | 310.4 | 57.4 | 387.4 | 46.9 | 435.2 |
| StepEntropy | 40.7 | 394.2 | 67.3 | 319.1 | 58.2 | 412.4 | 47.7 | 458.7 |
| Prune-on-Logic | 40.9 | **357.2** | 67.1 | 304.5 | 58.7 | 353.1 | 47.5 | 390.4 |
| GRACE | **42.0** | 377.1 | **67.7** | **288.7** | **59.3** | **327.5** | **48.2** | **354.1** |

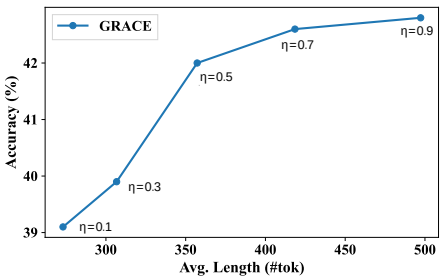



**Figure 3: Sensitivity Analysis of Compression Ratio $\eta$. Experiments are conducted on Mathverse.**

**Table 3: Evaluation of Visual Information Preservation.**

| Method | Visual Infor. ↑ |
|---|---|
| R1-Compress | 2.51 |
| Step Entropy | 2.79 |
| LIMOPro | 3.17 |
| Prune-on-Logic | 3.62 |
| GRACE | **3.78** |

All fine-tuning and post-training procedures are conducted using the LLaMA-Factory framework [61], which provides a unified and flexible interface for instruction tuning, reinforcement learning, and evaluation under diverse configurations.

Within the selector component of our framework, we incorporate a Graph Attention Network (GATv2) [2] to model the reasoning graph representations. This choice allows the model to capture fine-grained dependencies among multimodal reasoning nodes through adaptive attention weights, thereby enhancing the interpretability and expressiveness of the reasoning process. For hyperparameter settings, we fix the compression ratio to $\eta = 0.5$, controlling the degree of reasoning path reduction, and set the regularization coefficient to $\lambda = 1$, balancing the optimization between reasoning consistency and compression efficiency. For node-level text embedding, we employ Qwen3-Embedding-8B, a high-quality embedding model specifically optimized for the Qwen model family. This model provides semantically rich and well-aligned textual representations, which help maintain consistency between the selector's reasoning space and the underlying vision–language backbone.

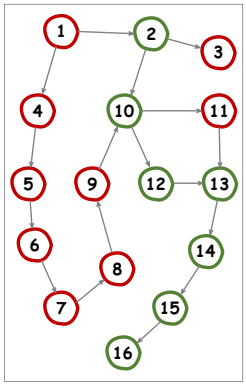
For quantitative assessment, we report both Accuracy and Average CoT length, which respectively reflect the effectiveness and efficiency of the reasoning process. Higher accuracy indicates better problem-solving capability, while a shorter average CoT length suggests more compact and efficient reasoning behaviors.

## 5.4 Overall Performance (RQ1)

As shown in Table 2, compared with No Compression, GRACE achieves comparable or even superior accuracy while significantly reducing token usage. On MMMU, the reasoning length decreases by 40.4% (from 594.2 to 354.1 tokens) with only a marginal 0.2% drop in accuracy. Notably, on MathVista, GRACE not only maintains the same level of accuracy as the full CoT model but slightly surpasses it (67.7% vs. 67.5%), demonstrating that appropriate reasoning compression can enhance both efficiency and reasoning quality. This indicates that compacting the CoT helps the model focus on the truly essential reasoning steps, thereby improving its ability to learn more concise and effective reasoning patterns.

Compared with Prune-on-Logic [60], which is designed for text-based reasoning compression, GRACE achieves markedly higher performance under similar or even shorter reasoning lengths. For instance, on MathVerse, GRACE outperforms Prune-on-Logic by 1.1 percentage points in accuracy while maintaining a shorter reasoning sequence. On MathVista, GRACE achieves both higher accuracy and lower token usage, suggesting that its learnable, graph-based selection mechanism effectively identifies critical reasoning steps and prunes redundant content. Benefiting from its learnable graph representation framework, GRACE preserves the essential logical structure of reasoning chains while removing minor or repetitive segments, achieving structure-aware CoT compression that balances interpretability and efficiency.

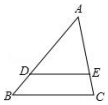
Compared with Chain-based CoT Compression methods (R1-Compression, LIMOPro, and StepEntropy), GRACE consistently achieves higher accuracy with shorter reasoning lengths across benchmarks. On MathVerse, GRACE outperforms the best among these baselines (StepEntropy, 40.7%) by 1.37% in accuracy while using less tokens (377.1 vs. 394.2). On MathVista, GRACE surpasses all baselines in accuracy and achieves the shortest CoT length (288.7 tokens), substantially improving over R1-Compress (310.4) and LIMOPro (329.8). On MMMU, GRACE further reduces token consumption by 18.6% compared to the most concise baseline (R1-Compress, 435.2 tokens), demonstrating its effectiveness.




**CoT:**
(1) Okay, so I'm trying to solve this geometry problem about triangle ABC where DE is parallel to BC. (2) The ratio of AD to AB is 3:4, and AC is 8 units long. (3) I need to find the length of AE, which is labeled as u. (4) Hmm, first, since DE is parallel to BC, the image shows that triangles ADE and ABC are similar. (5) This is because if two lines are parallel, the corresponding angles should be equal, making the triangles similar by AA (Angle-Angle) similarity. (6) So, if they're similar, their sides should be proportional. (7) That means the ratio of AD to AB should be the same as the ratio of AE to AC. (8) Let me write that down: AD / AB = AE / AC. (9) Given that AD/AB is 3/4, and AC is 8 units. (10) Plugging those into the equation: 3 / 4 = u / 8. (11) Now, I need to solve for u. (12) Cross-multiplying gives me: 4 * u = 3 * 8. (13) Calculating the right side: 4u = 24. (14) Then, dividing both sides by 4: u = 24 / 4 = 6. (15) So, u is 6.0 units. (16) Looking back at the choices, option C is 6.0, so that must be the correct answer.

**Question:**
In the provided diagram of triangle ABC, where line DE is parallel to line BC, the ratio of AD to AB is 3.0:4.0 and AC measures 8.0 units. What is the value of the variable u, representing the length of segment AE?

A: 8.0
B: 19.0
C: 6.0
D: 5.0




**Compressed Rationale:**
The ratio of AD to AB is 3:4, and AC is 8 units long. Plugging those into the equation: 3 / 4 = u / 8. Cross-multiplying gives me: 4 * u = 3 * 8. Calculating the right side: 4u = 24. Then, dividing both sides by 4: u = 24 / 4 = 6. So, u is 6.0 units. Looking back at the choices, option C is 6.0, so that must be the correct answer.

**Figure 4: Reasoning graph constructed by GRACE, along with the corresponding CoT and compressed rationale. Red nodes/text indicate the removed content, while green nodes/text represent the retained content.**

## 5.5 Sensitivity Analysis of Compression Ratio (RQ2)

In this section, we examine the impact of the compression ratio $\eta$ on model accuracy and reasoning efficiency. As illustrated in Figure 3, both accuracy and average CoT length decrease as $\eta$ decreases. This trend suggests that overly aggressive compression may cause the model to learn shorter reasoning habits, resulting in the omission of key intermediate steps necessary for complete inference. Excessive pruning may thus remove essential reasoning structures, leading to a decline in performance.

However, moderate compression yields a favorable trade-off between efficiency and accuracy. The elbow point of the performance curve appears around $\eta = 0.5$, indicating that this setting achieves a balance between preserving reasoning fidelity and improving inference efficiency. Consequently, we adopt $\eta = 0.5$ as the default configuration in all main experiments.

## 5.6 Analysis of Reasoning Graphs (RQ3)

To better understand how GRACE performs structured CoT compression, we conduct a qualitative analysis of the reasoning graphs it constructs, visualizing a representative sample from the GRACE-Dataset and examining which nodes are preserved or pruned.

As shown in Figure 4, GRACE constructs a directed graph where nodes represent reasoning steps and edges encode logical or semantic dependencies. During compression, it retains nodes carrying crucial deductive content (e.g., "Plugging those into the equation: 3/4=u/8") or pivotal transitions, while pruning redundant repetitions that merely restate known facts or intermediate results. The preserved nodes are typically semantically dense, causally connected, and located near key decision points, whereas pruned nodes often contain paraphrased explanations, unnecessary reflections, or repetitive visual cues that contribute little to the final answer. In the example, GRACE removes an entire branch containing repeated reasoning steps, demonstrating its structure-aware capability.

## 5.7 Evaluation of Visual Information Preservation (RQ4)

As noted in Section 1, chain-based CoT compression methods struggle with ultra-long multimodal reasoning chains. In contrast, graph structures naturally capture inter-step dependencies, better preserving critical visual details and cross-modal alignment cues. To assess whether GRACE retains essential visual information in its compressed CoTs, we adopt an "LLM-as-Judge" protocol using Qwen3-VL-8B. The evaluator scores each compressed CoT on a 1–5 scale based on two criteria: (i) reference to visually salient objects or fine-grained image details, and (ii) inclusion of explicit spatial or relational descriptions (prompt in Appendix ??). As shown in Table 3, GRACE outperforms all baselines, surpassing StepEntropy by 0.99 points. This gap indicates that text-only, step-level compression may discard vital visual grounding details when applied to multimodal reasoning. By explicitly modeling visual content during compression, GRACE preserves key cross-modal evidence, supporting more accurate and grounded multimodal reasoning.

## 5.8 Case Study on Fine-tuned Models (RQ5)

To further examine the impact of GRACE-compressed reasoning data on downstream multimodal reasoning models, we conduct a qualitative case study comparing models fine-tuned on the original long CoTs and those fine-tuned on GRACE-compressed CoTs.

As illustrated in Figure 5, the model fine-tuned on uncompressed CoTs (left, red box) tends to produce lengthy and repetitive reasoning traces. Although logically valid, these explanations often include redundant restatements of known conditions and unnecessary elaboration, leading to inefficient inference. In contrast, the model fine-tuned on GRACE-compressed data (right, green box) demonstrates a more concise and coherent reasoning process. It directly identifies the key geometric relations, infers the correct angle in fewer steps, and maintains logical completeness while using less than half the number of tokens. This observation highlights that GRACE not only enhances reasoning efficiency but also encourages models to internalize more compact and focused reasoning patterns. By filtering out redundant reasoning steps and retaining essential logical transitions, GRACE enables fine-tuned MLRMs to achieve more efficient inference without compromising accuracy.

## 6 Conclusion

In this paper, we proposed GRACE, a graph-based rationalized CoT compression framework designed to efficiently compress long reasoning chains in MLRMs. Specifically, we first converted the

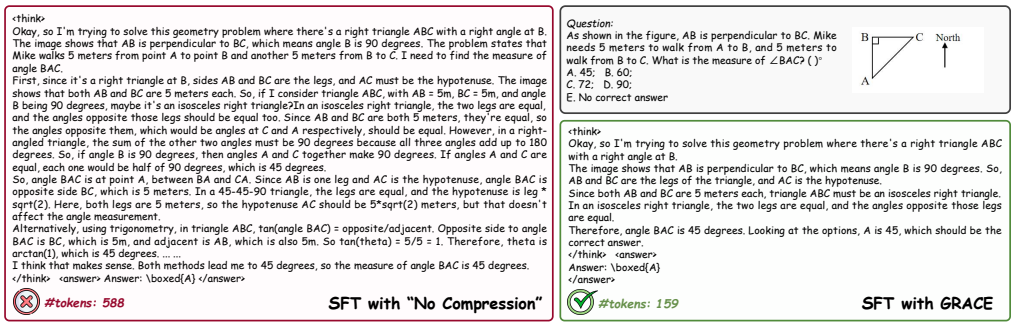



**Figure 5: Case study on fine-tuned models. The content in the red box on the left shows the output of the model fine-tuned on the original long CoT, while the content in the green box on the right shows the output of the model fine-tuned on GRACE-compressed reasoning data.**

original long CoTs into reasoning graphs. Then, a graph rationale compression framework, consisting of a selector and a predictor, was trained on these reasoning graphs. The selector learned to predict the importance of each node to the extract rationale subgraph, while the predictor used the selected rationale to generate the final prediction. After training, the trained selector was applied to uncompressed CoTs in new datasets to automatically extract key nodes and construct compressed subgraphs, effectively removing redundant reasoning steps. Finally, these compressed CoTs were subsequently used to build a SFT dataset, enabling MLLMs to retain essential reasoning information while significantly reducing redundancy. Extensive experiments on multiple multimodal benchmark datasets demonstrated that GRACE improved inference efficiency while consistently maintaining reasoning performance.

## References


[1] Shuai Bai, Keqin Chen, Xuejing Liu, et al. 2025. Qwen2.5-VL Technical Report. doi:10.48550/arXiv.2502.13923 arXiv:2502.13923 [cs].

[2] Shaked Brody, Uri Alon, and Eran Yahav. 2021. How attentive are graph attention networks? *arXiv preprint arXiv:2105.14491* (2021).

[3] Shiyu Chang, Yang Zhang, Mo Yu, and Tommi S. Jaakkola. 2020. Invariant Rationalization. In *Proceedings of the 37th International Conference on Machine Learning, (ICML)*.

[4] Liang Chen, Hongcheng Gao, Tianyu Liu, Zhiqi Huang, Flood Sung, Xinyu Zhou, Yuxin Wu, and Baobao Chang. 2025. G1: Bootstrapping Perception and Reasoning Abilities of Vision-Language Model via Reinforcement Learning. *arXiv preprint arXiv:2505.13426* (2025).

[5] Shuang Chen, Yue Guo, Zhaochen Su, Yafu Li, Yulun Wu, Jiacheng Chen, Jiayu Chen, Weijie Wang, Xiaoye Qu, and Yu Cheng. 2025. Advancing Multimodal Reasoning: From Optimized Cold Start to Staged Reinforcement Learning. *arXiv preprint arXiv:2506.04207* (2025).

[6] Xingyu Chen, Jiahao Xu, Tian Liang, Zhiwei He, Jianhui Pang, Dian Yu, Linfeng Song, Qiuzhi Liu, Mengfei Zhou, Zhuosheng Zhang, et al. 2024. Do not think that much for 2+ 3=? on the overthinking of o1-like llms. *arXiv preprint arXiv:2412.21187* (2024).

[7] Zhe Chen, Weiyun Wang, Yue Cao, Yangzhou Liu, Zhangwei Gao, Erfei Cui, Jinguo Zhu, Shenglong Ye, Hao Tian, Zhaoyang Liu, et al. 2024. Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. *arXiv preprint arXiv:2412.05271* (2024).

[8] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, and Peiyi et al. Wang. 2025. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. doi:10.48550/arXiv.2501.12948 arXiv:2501.12948 [cs].

[9] DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Chenggang Zhao, Chengqi Deng, and Chenyu et al. Zhang. 2025. DeepSeek-V3 Technical Report. doi:10.48550/arXiv.2412.19437 arXiv:2412.19437 [cs].

[10] Yihe Deng, Hritik Bansal, Fan Yin, Nanyun Peng, Wei Wang, and Kai-Wei Chang. 2025. Openvlthinker: An early exploration to complex vision-language reasoning via iterative self-improvement. *arXiv preprint arXiv:2503.17352* (2025).

[11] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, and Angela Fan. 2024. The llama 3 herd of models. *arXiv e-prints* (2024), arXiv–2407. https://ui.adsabs.harvard.edu/abs/2024arXiv240721783D/abstract

[12] Tingxu Han, Zhenting Wang, Chunrong Fang, Shiyu Zhao, Shiqing Ma, and Zhenyu Chen. 2024. Token-budget-aware llm reasoning. *arXiv preprint arXiv:2412.18547* (2024).

[13] Wenxuan Huang, Bohan Jia, Zijie Zhai, Shaosheng Cao, Zheyu Ye, Fei Zhao, Zhe Xu, Yao Hu, and Shaohui Lin. 2025. Vision-r1: Incentivizing reasoning capability in multimodal large language models. *arXiv preprint arXiv:2503.06749* (2025).

[14] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. 2024. Gpt-4o system card. *arXiv preprint arXiv:2410.21276* (2024).

[15] Eric Jang, Shixiang Gu, and Ben Poole. 2017. Categorical reparametrization with gumble-softmax. In *International Conference on Learning Representations (ICLR 2017)*. OpenReview. net.

[16] Gangwei Jiang, Yahui Liu, Zhaoyi Li, Qi Wang, Fuzheng Zhang, Linqi Song, Ying Wei, and Defu Lian. 2025. What Makes a Good Reasoning Chain? Uncovering Structural Patterns in Long Chain-of-Thought Reasoning. *arXiv preprint arXiv:2505.22148* (2025).

[17] KimiTeam, Angang Du, Bohong Yin, Bowei Xing, Bowen Qu, Bowen Wang, Cheng Chen, Chenlin Zhang, Chenzhuang Du, and Chu et al. Wei. 2025. Kimi-VL Technical Report. doi:10.48550/arXiv.2504.07491 arXiv:2504.07491 [cs].

[18] Ayeong Lee, Ethan Che, and Tianyi Peng. 2025. How well do llms compress their own chain-of-thought? a token complexity approach. *arXiv preprint arXiv:2503.01141* (2025).

[19] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. 2024. LLaVA-OneVision: Easy Visual Task Transfer. doi:10.48550/arXiv.2408.03326 arXiv:2408.03326 [cs].

[20] Zeju Li, Jianyuan Zhong, Ziyang Zheng, Xiangyu Wen, Zhijian Xu, Yingying Cheng, Fan Zhang, and Qiang Xu. 2025. Compressing chain-of-thought in llms via step entropy. *arXiv preprint arXiv:2508.03346* (2025).

[21] Weizhe Lin, Xing Li, Zhiyuan Yang, Xiaojin Fu, Hui-Ling Zhen, Yaoyuan Wang, Xianzhi Yu, Wulong Liu, Xiaosong Li, and Mingxuan Yuan. 2025. TrimR: Verifier-based Training-Free Thinking Compression for Efficient Test-Time Scaling. *arXiv preprint arXiv:2505.17155* (2025).

[22] Jinghui Lu, Haiyang Yu, Siliang Xu, Shiwei Ran, Guozhi Tang, Siqi Wang, Bin Shan, Teng Fu, Hao Feng, Jingqun Tang, et al. 2025. Prolonged Reasoning Is Not All You Need: Certainty-Based Adaptive Routing for Efficient LLM/MLLM Reasoning. *arXiv preprint arXiv:2505.15154* (2025).

[23] Pan Lu, Hritik Bansal, Tony Xia, Jiacheng Liu, Chunyuan Li, Hannaneh Hajishirzi, Hao Cheng, Kai-Wei Chang, Michel Galley, and Jianfeng Gao. 2023. Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts.

*arXiv preprint arXiv:2310.02255* (2023).

[24] Fanqing Meng, Lingxiao Du, Zongkai Liu, Zhixiang Zhou, Quanfeng Lu, Daocheng Fu, Botian Shi, Wenhai Wang, Junjun He, Kaipeng Zhang, et al. 2025. Mm-eureka: Exploring visual aha moment with rule-based large-scale reinforcement learning. *CoRR* (2025).

[25] Siqi Miao, Mia Liu, and Pan Li. 2022. Interpretable and generalizable graph learning via stochastic attention mechanism. In *International Conference on Machine Learning*. PMLR, 15524–15543.

[26] Sania Nayab, Giulio Rossolini, Marco Simoni, Andrea Saracino, Giorgio Buttazzo, Nicolamaria Manes, and Fabrizio Giacomelli. 2024. Concise thoughts: Impact of output length on llm reasoning and cost. *arXiv preprint arXiv:2407.19825* (2024).

[27] Yingzhe Peng, Gongrui Zhang, Miaosen Zhang, Zhiyuan You, Jie Liu, Qipeng Zhu, Kai Yang, Xingzhong Xu, Xin Geng, and Xu Yang. 2025. Lmm-r1: Empowering 3b lmms with strong reasoning abilities through two-stage rule-based rl. *arXiv preprint arXiv:2503.07536* (2025).

[28] Runqi Qiao, Qiuna Tan, Guanting Dong, Minhui Wu, Chong Sun, Xiaoshuai Song, Zhuoma GongQue, Shanglin Lei, Zhe Wei, Miaoxuan Zhang, et al. 2024. We-math: Does your large multimodal model achieve human-like mathematical reasoning? *arXiv preprint arXiv:2407.01284* (2024).

[29] Xiaoye Qu, Yafu Li, Zhaochen Su, Weigao Sun, Jianhao Yan, Dongrui Liu, Ganqu Cui, Daizong Liu, Shuxian Liang, Junxian He, et al. 2025. A survey of efficient reasoning for large reasoning models: Language, multimodality, and beyond. *arXiv preprint arXiv:2503.21614* (2025).

[30] QwenTeam. 2024. QVQ: To See the World with Wisdom. https://qwenlm.github.io/blog/qvq-72b-preview/

[31] Matthew Renze and Erhan Guven. 2024. The benefits of a concise chain of thought on problem-solving in large language models. In *2024 2nd International Conference on Foundation and Large Language Models (FLLM)*. IEEE, 476–483.

[32] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Y Wu, et al. 2024. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300* (2024).

[33] Haozhan Shen, Peng Liu, Jingcheng Li, Chunxin Fang, Yibo Ma, Jiajia Liao, Qiaoli Shen, Zilun Zhang, Kangjia Zhao, Qianqian Zhang, et al. 2025. Vlm-r1: A stable and generalizable r1-style large vision-language model. *arXiv preprint arXiv:2504.07615* (2025).

[34] StepFunTeam. [n. d.]. Step3: Cost-Effective Multimodal Intelligence. https://stepfun.ai/research/step3

[35] Yang Sui, Yu-Neng Chuang, Guanchu Wang, Jiamu Zhang, Tianyi Zhang, Jiayi Yuan, Hongyi Liu, Andrew Wen, Shaochen Zhong, Hanjie Chen, et al. 2025. Stop overthinking: A survey on efficient reasoning for large language models. *arXiv preprint arXiv:2503.16419* (2025).

[36] Huajie Tan, Yuheng Ji, Xiaoshuai Hao, Minglan Lin, Pengwei Wang, Zhongyuan Wang, and Shanghang Zhang. 2025. Reason-rft: Reinforcement fine-tuning for visual reasoning. *arXiv preprint arXiv:2503.20752* (2025).

[37] Kimi Team, Angang Du, Bofei Gao, Bowei Xing, Changjiu Jiang, Cheng Chen, Cheng Li, Chenjun Xiao, Chenzhuang Du, Chonghua Liao, et al. 2025. Kimi k1. 5: Scaling reinforcement learning with llms. *arXiv preprint arXiv:2501.12599* (2025).

[38] Peiyu Wang, Yichen Wei, Yi Peng, Xiaokun Wang, Weijie Qiu, Wei Shen, Tianyidan Xie, Jiangbo Pei, Jianhao Zhang, Yunzhuo Hao, et al. 2025. Skywork r1v2: Multimodal hybrid reinforcement learning for reasoning. *arXiv preprint arXiv:2504.16656* (2025).

[39] Weiyun Wang, Zhe Chen, Wenhai Wang, Yue Cao, Yangzhou Liu, Zhangwei Gao, Jinguo Zhu, Xizhou Zhu, Lewei Lu, Yu Qiao, and Jifeng Dai. 2025. Enhancing the Reasoning Ability of Multimodal Large Language Models via Mixed Preference Optimization. doi:10.48550/arXiv.2411.10442 arXiv:2411.10442 [cs].

[40] Yibo Wang, Haotian Luo, Huanjin Yao, Tiansheng Huang, Haiying He, Rui Liu, Naiqiang Tan, Jiaxing Huang, Xiaochun Cao, Dacheng Tao, et al. 2025. R1-compress: Long chain-of-thought compression via chunk compression and search. *arXiv preprint arXiv:2505.16838* (2025).

[41] Yibo Wang, Li Shen, Huanjin Yao, Tiansheng Huang, Rui Liu, Naiqiang Tan, Jiaxing Huang, Kai Zhang, and Dacheng Tao. 2025. R1-Compress: Long Chain-of-Thought Compression via Chunk Compression and Search. *arXiv preprint arXiv:2505.16838* (2025).

[42] Ying-Xin Wu, Xiang Wang, An Zhang, Xiangnan He, and Tat seng Chua. 2022. Discovering Invariant Rationales for Graph Neural Networks. In *ICLR*.

[43] Zhiyu Wu, Xiaokang Chen, Zizheng Pan, Xingchao Liu, et al. 2024. DeepSeek-VL2: Mixture-of-Experts Vision-Language Models for Advanced Multimodal Understanding. doi:10.48550/arXiv.2412.10302 arXiv:2412.10302 [cs].

[44] Heming Xia, Yongqi Li, Chak Tou Leong, Wenjie Wang, and Wenjie Li. 2025. Tokenskip: Controllable chain-of-thought compression in llms. *arXiv preprint arXiv:2502.12067* (2025).

[45] Yang Xiao, Jiashuo Wang, Ruifeng Yuan, Chunpu Xu, Kaishuai Xu, Wenjie Li, and Pengfei Liu. 2025. LIMOPro: Reasoning Refinement for Efficient and Effective Test-time Scaling. *arXiv preprint arXiv:2505.19187* (2025).

[46] Guowei Xu, Peng Jin, Ziang Wu, Hao Li, Yibing Song, Lichao Sun, and Li Yuan. 2025. LLaVA-CoT: Let Vision Language Models Reason Step-by-Step. doi:10.48550/arXiv.2411.10440 arXiv:2411.10440 [cs].

[47] Xiaoang Xu, Shuo Wang, Xu Han, Zhenghao Liu, Huijia Wu, Peipei Li, Zhiyuan Liu, Maosong Sun, and Zhaofeng He. 2025. A*-Thought: Efficient Reasoning via Bidirectional Compression for Low-Resource Settings. *arXiv preprint arXiv:2505.24550* (2025).

[48] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388* (2025).

[49] Yi Yang, Xiaoxuan He, Hongkun Pan, Xiyan Jiang, Yan Deng, Xingtao Yang, Haoyu Lu, Dacheng Yin, Fengyun Rao, Minfeng Zhu, et al. 2025. R1-onevision: Advancing generalized multimodal reasoning through cross-modal formalization. *arXiv preprint arXiv:2503.10615* (2025).

[50] Huanjin Yao, Jiaxing Huang, Wenhao Wu, Jingyi Zhang, Yibo Wang, Shunyu Liu, Yingjie Wang, Yuxin Song, Haocheng Feng, Li Shen, and Dacheng Tao. 2024. Mulberry: Empowering MLLM with o1-like Reasoning and Reflection via Collective Monte Carlo Tree Search. doi:10.48550/arXiv.2412.18319 arXiv:2412.18319 [cs].

[51] Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, Hongji Zhu, Tianchi Cai, Haoyu Li, Weilin Zhao, Zhihui He, Qianyu Chen, Huarong Zhou, Zhensheng Zou, Haoye Zhang, Shengding Hu, Zhi Zheng, Jie Zhou, Jie Cai, Xu Han, Guoyang Zeng, Dahai Li, Zhiyuan Liu, and Maosong Sun. 2024. MiniCPM-V: A GPT-4V Level MLLM on Your Phone. doi:10.48550/arXiv.2408.01800 arXiv:2408.01800 [cs].

[52] Hang Yuan, Bin Yu, Haotian Li, Shijun Yang, Christina Dan Wang, Zhou Yu, Xueyin Xu, Weizhen Qi, and Kai Chen. 2025. Not All Tokens Are What You Need In Thinking. *arXiv preprint arXiv:2505.17827* (2025).

[53] Linan Yue, Yichao Du, Yizhi Wang, Weibo Gao, Fangzhou Yao, Li Wang, Ye Liu, Ziyu Xu, Qi Liu, Shimin Di, et al. 2025. Don't Overthink It: A Survey of Efficient R1-style Large Reasoning Models. *arXiv preprint arXiv:2508.02120* (2025).

[54] Linan Yue, Qi Liu, Yichao Du, Yanqing An, Li Wang, and Enhong Chen. 2022. DARE: Disentanglement-Augmented Rationale Extraction. In *Advances in Neural Information Processing Systems*, Vol. 35. 26603–26617.

[55] Xiang Yue, Yuansheng Ni, Kai Zhang, Tianyu Zheng, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, et al. 2024. Mmmu: A massive multi-discipline multimodal understanding and reasoning benchmark for expert agi. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 9556–9567.

[56] Duzhen Zhang, Yahan Yu, Jiahua Dong, Chenxing Li, Dan Su, Chenhui Chu, and Dong Yu. 2024. Mm-llms: Recent advances in multimodal large language models. *arXiv preprint arXiv:2401.13601* (2024).

[57] Haotian Zhang, Mingfei Gao, Zhe Gan, Philipp Dufter, Nina Wenzel, Forrest Huang, Dhruti Shah, Xianzhi Du, Bowen Zhang, Yanghao Li, Sam Dodge, Keen You, Zhen Yang, Aleksei Timofeev, Mingze Xu, Hong-You Chen, Jean-Philippe Fauconnier, Zhengfeng Lai, Haoxuan You, Zirui Wang, Afshin Dehghan, Peter Grasch, and Yinfei Yang. 2024. MM1.5: Methods, Analysis & Insights from Multimodal LLM Fine-tuning. doi:10.48550/arXiv.2409.20566 arXiv:2409.20566 [cs].

[58] Jingyi Zhang, Jiaxing Huang, Huanjin Yao, Shunyu Liu, Xikun Zhang, Shijian Lu, and Dacheng Tao. 2025. R1-VL: Learning to Reason with Multimodal Large Language Models via Step-wise Group Relative Policy Optimization. doi:10.48550/arXiv.2503.12937 arXiv:2503.12937 [cs].

[59] Renrui Zhang, Dongzhi Jiang, Yichi Zhang, Haokun Lin, Ziyu Guo, Pengshuo Qiu, Aojun Zhou, Pan Lu, Kai-Wei Chang, Yu Qiao, et al. 2024. Mathverse: Does your multi-modal llm truly see the diagrams in visual math problems?. In *European Conference on Computer Vision*. Springer, 169–186.

[60] Shangziqi Zhao, Jiahao Yuan, Guisong Yang, and Usman Naseem. 2025. Can Pruning Improve Reasoning? Revisiting Long-CoT Compression with Capability in Mind for Better Reasoning. *arXiv preprint arXiv:2505.14582* (2025).

[61] Yaowei Zheng, Richong Zhang, Junhao Zhang, Yanhan Ye, Zheyan Luo, Zhangchi Feng, and Yongqiang Ma. 2024. LlamaFactory: Unified Efficient Fine-Tuning of 100+ Language Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*. Association for Computational Linguistics, Bangkok, Thailand. http://arxiv.org/abs/2403.13372

[62] Hengguang Zhou, Xirui Li, Ruochen Wang, Minhao Cheng, Tianyi Zhou, and Cho-Jui Hsieh. 2025. R1-Zero's" Aha Moment" in Visual Reasoning on a 2B Non-SFT Model. *arXiv preprint arXiv:2503.05132* (2025).

[63] Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, et al. 2025. Internvl3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479* (2025).

[64] Ren Zhuang, Ben Wang, and Shuifa Sun. 2025. Accelerating Chain-of-Thought Reasoning: When Goal-Gradient Importance Meets Dynamic Skipping. *arXiv preprint arXiv:2505.08392* (2025).

## A Variant Experiments

To validate the effectiveness of GRACE's graph construction mechanism, we compare it against two variants: (i) Random, where edges are assigned randomly, and (ii) Fully-Connected, where every reasoning step is connected to all others. As shown in Table 4, GRACE consistently outperforms both variants across all three benchmarks in both accuracy and average CoT length. The gains demonstrate that explicitly modeling semantic dependencies between reasoning steps, rather than using arbitrary or dense connections, is crucial for effective compression and performance retention.

## B Generalization Across MLLM Backbones

To assess GRACE's compatibility with different MLLMs, we examine its performance under various backbone models. While our main experiments are based on Qwen-series, we additionally evaluate GRACE on InternVL3-8B [63]. Results in Table 5 show that GRACE consistently improves inference efficiency while maintaining or slightly enhancing accuracy compared to both the uncompressed baseline and a strong pruning baseline (Prune-on-Logic). These results confirm that GRACE's graph-based compression framework generalizes well beyond Qwen Series, highlighting its broad applicability across different MLLMs.

## C Prompt for Text Smoothing

**Prompt for Text Smoothing**

### Task ###
Combine the given reasoning steps into a single, fluent Chain-of-Thought. - Preserve the exact content and logical order of all steps.
- Make only minimal edits (e.g., add or adjust linking words like "then", "since", "therefore", or punctuation) to ensure smooth transitions.
- Do not omit, rephrase, or alter any substantive reasoning content.
- If the sequence of steps is already fluent and well-connected, output it exactly as-is without any modification.

### Examples ###
{Examples}

## D Prompt for Node Splitting

**Prompt for Node Splitting ($P_{\text{node}}$)**

### Role ###
You are a world-class reasoning structure parser with expertise in logic, mathematics, and discourse analysis. Your job is to dissect Chain-of-Thought (CoT) reasoning passages into minimal, semantically complete reasoning units - each representing one atomic step in the logical flow.
### Task ###
Given a problem-solving passage, segment it into discrete reasoning trunks. Each trunk must:
- Represent ONE atomic reasoning action (e.g., define, compute, compare, infer, assume, conclude).
- Be semantically self-contained - understandable without needing adjacent trunks.
- Preferably 1 sentence; up to 2 sentences MAXIMUM. Never exceed 3.
- Preserve the original wording - DO NOT paraphrase or summarize.
- NEVER split across causal/logical connectors (e.g., "because...so...", "if...then...", "since...therefore...") - keep them together.
- ALWAYS include the final answer (especially if boxed) as a separate trunk.
### Splitting Rules ###
SPLIT when:
- A new reasoning action begins (e.g., after "Next, ...", "Then, ...")
- A new variable/formula is introduced
- A conclusion is drawn from prior steps
DO NOT SPLIT:
- Across causal/conditional phrases ("because...", "since...", "if...then...")
- Mathematical expressions (keep "x = a + b ⟶ x = 5" intact)
### Output Format ###
Strictly output in this XML-style format. One trunk per line. Preserve original order.
<node>[Exact text from CoT]</node>
Preserve original sentence order. Strictly follow the format - NO explanations, NO extra text, NO numbering. ONLY nodes in specified format.

### Examples ###
{Examples}

## E Prompt for Edge Extraction

**Prompt for Edge Construction ($P_{\text{edge}}$)**

### Role ###
You are a reasoning graph architect specializing in DAG (Directed Acyclic Graph) construction. Your task is to build accurate, cycle-free dependency edges between segmented reasoning nodes - capturing branching, merging, and multi-premise logic while strictly forbidding cycles or backward edges.

### Task ###
Given a numbered list of <node> elements, assign directed edges from earlier nodes to later nodes based on logical, mathematical, or semantic dependency. Each edge must reflect how a later node builds upon, references, or branches from one or more prior nodes. The resulting graph must be a valid DAG - no cycles, no backward edges.

### Rules ###
– Forward-Only Edges: An edge may only go from Node A to Node B if A's id < B's id.

**Table 4: Performance comparison on variants of GRACE. Accuracy (Acc.) measures effectiveness, while average CoT length (Avg. length) reflects reasoning efficiency.**

| Method | MathVerse | | MathVista | | WeMath | |
|---|---|---|---|---|---|---|
| | Acc. (%) ↑ | Avg. length (#tok) ↓ | Acc. (%) ↑ | Avg. length (#tok) ↓ | Acc. (%) ↑ | Avg. length (#tok) ↓ |
| Random | 31.7 | 385.2 | 55.1 | 297.8 | 48.2 | 395.4 |
| Fully-Connected | 39.8 | 410.5 | 64.2 | 329.6 | 54.9 | 437.1 |
| GRACE | 42.0 | 377.1 | 67.7 | 288.7 | 59.5 | 369.3 |

**Table 5: Results with InternVL3-8B as backbone. Accuracy (Acc.) measures effectiveness, while average CoT length (Avg. length) reflects reasoning efficiency.**

| InternVL3-8B as backbone | MathVerse | | MathVista | | WeMath | |
|---|---|---|---|---|---|---|
| | Acc. (%) ↑ | Avg. length (#tok) ↓ | Acc. (%) ↑ | Avg. length (#tok) ↓ | Acc. (%) ↑ | Avg. length (#tok) ↓ |
| No-Compression | 48.5 | 557.6 | 57.1 | 346.1 | 60.9 | 489.0 |
| Prune-on-Logic | 47.5 | 449.1 | 56.3 | 319.2 | 59.8 | 407.3 |
| GRACE | 47.7 | 412.2 | 56.8 | 267.4 | 60.4 | 385.5 |

– No Cycles Allowed: Under no circumstance create a circular dependency.
– Graph Richness Requirement: Actively construct the most semantically rich and densely connected graph possible. Connect nodes wherever logically justified - even if multiple edges, multi-parent or multi-child nodes result. DO NOT output a trivial linear chain (e.g., 1->2->3->4...) unless the reasoning is genuinely devoid of computation, logic, or reflection. Enrichment over simplification.
... ...
– Support Rich Structures:
– Multiple parents -> allowed (e.g., Node5 depends on Node2 AND Node3)
– Branching paths -> use alternative_of for conditional or assumption splits
– Converging conclusions -> allowed (e.g., summary node depends on multiple results)
– Justification Requirement:
– Quote exact phrase from node text that implies the dependency
– Concisely describe the logical bridge (e.g., "uses variable x from Node 1")
– DO NOT invent content. DO NOT paraphrase unnecessarily.
– Edge Types (choose one per edge):
– computes_from - B performs a mathematical, symbolic, or formula-based derivation that directly uses A's output, value, or defined entity.
– depends_on - B logically or semantically requires A as a premise, condition, definition, or supporting fact - but NOT via computation.

– sequential - B follows A in narrative, procedural, or discourse order, but shares NO computational, logical, or reflective dependency.
– reflects - B represents the **initiating act** of metacognitive reflection: checking, verifying, doubting, or reconsidering the validity, consistency, or interpretation of A's content.
– branches_from - B introduces a new assumption, hypothetical path, or alternative interpretation that diverges from A - but is NOT triggered by reflection or doubt.
– Uncertainty Handling: If no clear dependency exists, DO NOT create an edge. It's better to under-connect than to risk cycles or spurious links.
### Output Format ###
Strictly output in the following format. One edge per line.

<edge from=N to=M type=EDGE_TYPE>Justification: QUOTE OR LOGIC</edge>

-> NO extra text. NO explanations. ONLY the above.

### Examples ### {Examples}